ASHISH TAMHANKAR, Ph.D.

Computational Chemist | Computer-Aided Drug Design | Molecular Modelling| ML

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PROFESSIONAL SUMMARY

Computational and bioinorganic chemist with 4+ years of research experience spanning structure-based drug design, molecular dynamics/docking of membrane-protein targets, and quantum chemical modelling of metalloenzyme reactivity, with projects ranging from A3 adenosine G-protein-coupled receptor (GPCR) ligand recognition to copper-dependent lytic polysaccharide monooxygenases (LPMOs) and synthetic diiron peroxo complexes. Lead computational chemist within the multi-institutional ERC Synergy Grant CUBE consortium (Germany/Norway/Italy), with a record of peer-reviewed publications, conference presentations, and mentoring junior scientists. Enthusiastic to learn and apply quantum algorithms or quantum computing technologies to drug discovery problems.

CORE EXPERTISE

- **Computer-aided drug design:** Molecular dynamics, molecular docking, free-energy perturbation (FEP), Structure-based Drug Design using virtual screening for membrane-protein (GPCR) systems.
- **Molecular property prediction and optimization:** QresFEP-based analysis of mutational data to prioritize and optimize candidate molecules based on predicted binding affinity.
- **Computational chemistry:** DFT, TDDFT, and QM/MM calculations to understand structure–function relationships relevant to molecular design and reactivity of metalloenzymes.
- **Programming & data analysis:** Design and automation of reproducible, large-scale computational campaigns on HPC/SLURM systems using Python and Bash; scientific machine learning (supervised learning, model validation); Python pipelines (developed with AI-assisted coding tools) for processing spectroscopic, kinetic, mutagenetic, and computational datasets.
- **Software:** ORCA, GROMACS, AMBER, Schrödinger Suite (Maestro, Glide, FEP+), xTB, VMD, PyMOL, RDKit, AlphaFold, Python (NumPy, SciPy, scikit-learn, pandas).
- **Collaboration & leadership:** Lead computational chemist within CUBE consortium. Integrated computations with experimental X-ray spectroscopy and protein reaction kinetic data across multi-institutional collaborations; mentoring of undergraduate researchers; scientific trainer for reactivity/spectroscopy practicals; elected PhD representative.
- **Communication & outreach:** Oral and poster presentations at international conferences; co-organizer of a student-led international conference; collaborative manuscript preparation across multi-group projects; presented at annual and bi-weekly CUBE consortium and DeBeer-Neese research symposium meetings providing project updates.

RESEARCH EXPERIENCE

Postdoctoral Researcher (Guest Contract) | Max Planck Institute for Chemical Energy Conversion May 2026 - Present

- Group of Prof. Dr. Serena DeBeer, Department of Inorganic Spectroscopy, Mülheim an der Ruhr, Germany
- Continuing collaborative research within the ERC Synergy Grant CUBE consortium, finalizing manuscripts on second-sphere residue effects and substrate-binding modulation in lytic polysaccharide monooxygenases (LPMOs).

Ph.D. Researcher | Max Planck Institute for Chemical Energy Conversion Jul 2022 - May 2026

- Group of Prof. Dr. Serena DeBeer, Department of Inorganic Spectroscopy, Mülheim an der Ruhr, Germany
- **Consortium research:** Led two collaborative biocatalysis projects on LPMOs within the CUBE consortium.
- **Metalloenzyme mechanism (LPMOs):** Used QM/MM (DFT) and Local Energy Decomposition (LED) calculations to dissect how second-sphere residues modulate O_2 and H_2O_2 activation in lytic polysaccharide monooxygenases (two manuscripts in preparation).
- **Electronic structure & spectroscopy integration:** Combined high-energy-resolution X-ray absorption spectroscopy (HERFD-XAS) with electronic structure calculations (TDDFT) to dissect first- and second-sphere effects in Cu(I) LPMOs (manuscript submitted, *Inorg. Chem.*); supported interpretation of Cu K-edge XAS data for multiple collaborative LPMOs (*Inorg. Chem.* 2024).
- **Model complex reactivity:** Contributed computational and electronic-structure analysis to studies of diiron complex's formation pathways and reactivity (*JACS* 2026).
- **Structural/electronic modulation studies:** Investigated structural and electronic changes in LPMOs upon chitin binding using combined computations, binding free energy, and reactivity combined with experimental data (*Chem. Sci.* 2025); identified a lysine–cysteine redox switch with a NOS bridge in transaldolase via sulfur K-edge XAS and TDDFT calculations (*J. Phys. Chem. Lett.* 2024).
- **Workflow development:** Designed and automated reproducible computational pipelines (QM/MM, DFT, free-energy profiling) on HPC systems using Python, Bash, and SLURM; built Python-based workflows to process and analyze large computational and spectroscopic datasets.

Research Intern | Uppsala University Sep 2020 - Nov 2021

- Group of Prof. Dr. Hugo Gutiérrez-de Terán, Department of Cell and Molecular Biology, Uppsala, Sweden
- Performed molecular docking and molecular dynamics simulations to investigate protein–ligand interactions on a series of A3 adenosine GPCR antagonists (*ACS Med. Chem. Lett.* 2022).
- Carried out free-energy perturbation (FEP) calculations and QresFEP-based analysis of GPCRdb mutational data to characterize the impact of mutations on ligand binding.
- Designed computationally informed mutations to probe binding modes and protein–ligand interactions, and the impact of ligand binding to develop novel pharmacological tools for A3 GPCR antagonists. (*M.Sc. thesis*)

EDUCATION

Ph.D. in Chemistry (Computational Biochemistry)| Max Planck Institute for Chemical Energy Conversion / Ruhr University Bochum
Advisor: Prof. Dr. Serena DeBeer Jul 2022 - May 2026
Dissertation: *"Quantum Chemical Investigations of Bioinorganic Reactivity and Spectroscopy in Biological Systems and Model Complexes."*
Magna Cum Laude (sehr gut)

M.Sc. in Pharmaceutical Sciences – Drug Discovery and Development | Uppsala University
Graduated with Distinction (VG) for thesis. Sep 2019 - Jun 2021
Advisor: Prof. Dr. Hugo Gutiérrez-de Terán
Thesis: *"Structure-Based Design of Novel Pharmacological Tools for the A3 Adenosine G Protein-Coupled Receptor (GPCR)."*
Coursework: Computational Medicinal Chemistry; Advanced Molecular Modelling Applied to Drug Discovery; Pharmaceutical Bioinformatics and Sequence Analysis; Models for Biological Systems.

B.Sc. in Pharmacy | University of Pune
Graduated with First Class. Aug 2014 - May 2018
Coursework: Medicinal Chemistry; Pharmacology; Pharmaceutical Biochemistry.

AWARDS

Best Poster, DeBeer–Neese Research Summer Symposium 2025
Best Poster, DeBeer–Neese Research Winter Symposium 2024
Best Poster, DeBeer–Neese Research Summer Symposium 2024

TEACHING AND MENTORING

Tutor | Max Planck Institute for Chemical Energy Conversion 2024 - 2026

- Mentored two Bachelor's students, designing QM/MM calculations to identify mutations with the greatest impact on LPMO catalytic rates; XES calculations on a series of diiron complexes.

Scientific Trainer | Max Planck Institute 2023 - 2025

- Max Planck International School on Spectroscopy and Electronic Structure of Transition Metal Complexes (~50 undergraduate and graduate students/year).
- Assisted in developing course material for "Excited State Spectroscopy" practical session and instructed and assisted students during practical session, including using ORCA computational chemistry package.

SERVICE AND OUTREACH

- Organizer, StuCon-2025 (Sponsorship Team), co-organized a student-led international conference hosted jointly by MPI für Kohlenforschung and MPI for Chemical Energy Conversion, 2025.
- External Ph.D. Representative, Max Planck Institute for Chemical Energy Conversion, 2024 - 2025.
- Organized CUBE consortium bi-weekly meetings, 2025-2026.
- Attended and presented a poster at Max Planck International School on Spectroscopy and Electronic Structure of Transition Metal Complexes, 2023.
- Attended and presented a poster at Bioexcel CSC Spring School on Computational Chemistry, 2025.

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Cyclic Peptide Modelling and Informatics, Schrödinger, 2026.
- Applications of Free Energy Calculations in Modern Drug Hunting, Schrödinger, 2026.
- Applied Machine Learning with Python, Coursera, 2022.
- Introduction to Data Science with Python, Coursera, 2022.
- Computer-Aided Drug Design, Rasa Life Science Informatics, India, 2018.
- Drug Development, Coursera, 2018.

PUBLICATIONS

In preparation

[9] Tamhankar, A.; Rønnekleiv, S. E.; Forsberg, Z.; Røhr, Å. K.; Eijsink, V. G. H.; DeBeer, S.; Jannuzzi, S. A. V. Cooperative Interplay between Second Sphere Residues Modulates H_2O_2 Production from O_2 in Lytic Polysaccharide Monooxygenases. *In preparation* **2026.**

[8] Tamhankar, A.; Lee, H.; Rønnekleiv, S. E.; Røhr, Å. K.; Eijsink, V. G. H.; Jannuzzi, S. A. V.; DeBeer, S. The Role of the Third Histidine Residue in Lytic Polysaccharide Monooxygenases in H_2O_2 Activation: Insights from QM/MM Calculations. *In preparation* **2026.**

Submitted

[7] Tamhankar, A.; Joseph, C.; Jannuzzi, S. A. V.; Hall, K. R.; Munzone, A.; Sørlie, M.; Eijsink, V. G. H.; Orio, M.; Simaan, A. J.; Decroos, C.; DeBeer, S. Dissecting First- and Second-Sphere Effects in Cu(I) Lytic Polysaccharide Monooxygenases by High-Energy Resolution X-ray Absorption Spectroscopy and Electronic Structure Calculations. *Submitted manuscript (Inorg. Chem.)* **2026.**

Peer-reviewed

[6] Jansen, S.; Röhs, F. L. B.; Zimmermann, T. P.; Walleck, S.; van Gastel, M.; **Tamhankar, A.**; Zhang, B.; Jannuzzi, S. A. V.; Kellner, I.; Ivanović-Burmazović, I.; DeBeer, S.; Glaser, T. Formation Pathways of an Electrophilic μ-1,2-Peroxo Fe(III)Fe(III) Complex: Spectroscopic Characterization and Reactivity. *Journal of the American Chemical Society* 2026, *148* (7), 7065–7083. https://doi.org/10.1021/jacs.5c18121.

[5] Joseph, C.; **Tamhankar, A.**; Golten, O.; Sengupta, K.; Jannuzzi, S. A. V.; Sørlie, M.; Kang, L.; Røhr, Å. K.; Eijsink, V. G. H.; DeBeer, S. Structural and Electronic Modulations of Lytic Polysaccharide Monooxygenase (LPMO) upon Chitin Binding: Insights from X-ray Spectroscopy. *Chemical Science* 2025, *16*, 22952–22969. https://doi.org/10.1039/D5SC07620J.

[4] Munzone, A.; Pujol, M.; **Tamhankar, A.**; Joseph, C.; Mazurenko, I.; Réglier, M.; Jannuzzi, S. A. V.; Royant, A.; Sicoli, G.; DeBeer, S.; Orio, M.; Simaan, A. J.; Decroos, C. Integrated Experimental and Theoretical Investigation of Copper Active Site Properties of a Lytic Polysaccharide Monooxygenase from *Serratia marcescens*. *Inorganic Chemistry* 2024, *63* (24), 11063–11078. https://doi.org/10.1021/acs.inorgchem.4c00602.

[3] Tamhankar, A.; Wensien, M.; Jannuzzi, S. A. V.; Chatterjee, S.; Lassalle-Kaiser, B.; Tittmann, K.; DeBeer, S. In Solution Identification of the Lysine-Cysteine Redox Switch with a NOS Bridge in Transaldolase by Sulfur K-Edge X-Ray Absorption Spectroscopy. *Journal of Physical Chemistry Letters* 2024, *15* (16), 4263–4267. https://doi.org/10.1021/acs.jpclett.4c00484.

[2] Hall, K. R.; Joseph, C.; Ayuso-Fernández, I.; **Tamhankar, A.**; Rieder, L.; Skaali, R.; Golten, O.; Neese, F.; Røhr, Å. K.; Jannuzzi, S. A. V.; DeBeer, S.; Eijsink, V. G. H.; Sørlie, M. A Conserved Second Sphere Residue Tunes Copper Site Reactivity in Lytic Polysaccharide Monooxygenases. *Journal of the American Chemical Society* 2023, *145* (34), 18888–18903. https://doi.org/10.1021/jacs.3c05342.

[1] Miranda-Pastoriza, D.; Bernárdez, R.; Azuaje, J.; Prieto-Díaz, R.; Majellaro, M.; **Tamhankar, A. V.**; Koenekoop, L.; González, A.; Gioé-Gallo, C.; Mallo-Abreu, A.; Brea, J.; Loza, M. I.; García-Rey, A.; García-Mera, X.; Gutiérrez-de Terán, H.; Sotelo, E. Exploring Non-Orthosteric Interactions with a Series of Potent and Selective A3 Antagonists. *ACS Medicinal Chemistry Letters* 2022, *13* (2), 243–249. https://doi.org/10.1021/acsmedchemlett.1c00598.

SELECTED PRESENTATIONS

Oral

- International Conference on Coordination Chemistry (ICCC 46), 2026, Odense, Denmark: "Coupled Second Sphere Residues Boost H_2O_2 Production in Lytic Polysaccharide Monooxygenases."
- Invited speaker, Group of Prof. Dr. Erna Wieduwilt, University of Duisburg-Essen, Germany, 2026: “Computational Studies of Lytic Polysaccharide Monooxygenases: Secondary Sphere Mutations and Substrate Effects.”
- LUCIA A 20 ANS, SOLEIL Synchrotron, 2024, Saint-Aubin, France: "In Solution Identification of the Lysine-Cysteine Redox Switch with a NOS Bridge in Transaldolase by Sulfur K-Edge X-Ray Absorption Spectroscopy."
- CUBE consortium annual meetings; 2024, 2025: “Computational Insights in LPMO Reactivity.”

Poster

- CSC Spring School on Computational Chemistry, 2025, Espoo, Finland: "Synergy between Second Sphere Residues Boosts H_2O_2 Production from O_2 in Lytic Polysaccharide Monooxygenases: A QM/MM Study."
- The 4th LPMO Symposium, 2024, York, UK: "Computational Insights into the Impact of Chitin Binding by Lytic Polysaccharide Monooxygenase with X-ray Spectroscopy."
- 17th European Biological Inorganic Chemistry Conference (EuroBIC-17), 2024, Münster, Germany: "Computational Insights into the Impact of Chitin Binding by Lytic Polysaccharide Monooxygenase with X-ray Spectroscopy."

- 6th Quantum Bio-Inorganic Chemistry Conference (QBIC-VI), 2023, Warsaw, Poland: "Second Sphere Residue Effects on K-edge X-ray Absorption Spectroscopy and Reactivity in Lytic Polysaccharide Monooxygenases."

LANGUAGES

- English (fluent); German (A2); Swedish (basic); Marathi and Hindi (native)

REFERENCES

- Prof. Dr. Serena DeBeer, Max Planck Institute for Chemical Energy Conversion, serena.debeer@cec.mpg.de
- Prof. Dr. Sergio A. V. Jannuzzi, Max Planck Institute for Chemical Energy Conversion, sergio.jannuzzi@cec.mpg.de
- Prof. Hugo Gutiérrez-de Terán, Uppsala University, h.g.teran@cinn.es